

Multi-Drug Response Prediction & Optimization for Selective Combination Therapy

Islam Tayeb* Bo Chi*

*Duke University

The Opportunity

Treating patients with multiple drugs administered concurrently—combination therapy—is standard for serious diseases, but drug interactions produce non-additive toxicities and synergies. Exhaustive pairwise screening costs millions of dollars per cellular context and takes weeks [1]. Existing computational approaches are insufficient: Bulk synergy models (DeepSynergy, MARSY) learn identifiers instead of molecular features [2–4]; single-cell perturbation models (scGen, GEARS) remain limited to genetic perturbations or single drugs [5–7]. Arc Institute’s STATE [8] conditions on cell state and drug identity to predict post-perturbation expression; it generalizes across cell types, performing best on known drugs in unseen contexts [9]. We extend STATE with **continuous drug representations** and **multi-objective optimization** for selective combination therapy.

Research Idea & Approach

Aim #1: Extend State to multi-drug, multi-objective optimization. STATE sums a drug embedding with cell-state embeddings before passing into a transformer that predicts post-perturbation expression. Its differentiable forward pass lets us directly optimize drug inputs, finding effective combinations across cell types. We plan to add:

1. **Continuous drug encoder.** Target-based embeddings (representing drugs by their protein targets) significantly outperform molecular-structure models for chemical perturbation prediction [9], yet STATE uses one-hot drug indices carrying no structural information [8]. We replace these with target-based embeddings projected into STATE’s embedding space, benchmarking against molecular fingerprints as a fallback representation.
2. **Additive drug composition.** For a combination, we sum individual single-drug predictions conditioned on the same cell context. This additive approach is the strongest known baseline for genetic double perturbations [7].
3. **Multi-objective optimization.** We backpropagate through STATE in continuous embedding space, maximizing target-cell efficacy while penalizing healthy-cell expression changes, and retrieve the nearest real compounds. We benchmark against single-objective optimization and random search, testing whether the optimizer recovers known selective regimens (e.g., combinations from NCI-ALMANAC [10] with documented cell-line-selective synergy).

Aim #2: Predict non-additive drug interactions. Aim #1 assumes additivity. While additive baselines outperform all published deep learning methods for genetic double perturbations [7], drug combinations provide richer supervision (dose-response matrices, null models across thousands of pairs [10, 11]) that may supply the signal non-additive models need. Two considerations motivate this direction:

1. **The additivity finding is for genetic perturbations.** Datasets showing additive dominance were purely CRISPR-based; drug combinations differ through non-linear dose-response, target competition, and pharmacokinetic interactions [12]. Whether additivity holds for chemical combinations at single-cell resolution is untested.
2. **State’s distributional loss avoids mode collapse.** Point-wise losses drive non-additive models to collapse to additive predictions, because the population mean that dominates these objectives is approximately additive for most genes [7, 13]. STATE leverages a distributional (maximum mean discrepancy) loss [8] across the full cell distributions instead, avoiding mean collapse.

To capture these interactions, we replace Aim #1’s additive sum with a self-attention encoder over drug embeddings that outputs a single combination representation, capturing cross-drug modulation on top of single contributions. We train this module to predict the residual between observed combination effects and the additive baseline, using combination transcriptomics [14, 15] with bulk synergy scores [10, 11] as auxiliary supervision.

We evaluate on leave-drug-out and leave-cell-line-out splits against naive additivity, CPA [14], and MARSY [3]. If interaction modules beat additive baselines, the result has direct implications for combination screening. Otherwise, the first single-cell additivity benchmark for drug combinations is itself novel.

Training Goals

We have published in computational biology and built production systems at YC-backed startups, but separately: the research stayed in papers, and the production work was not science. This fellowship allows us to build scientifically grounded tools that affect real users. Arc’s vertical integration of Tahoe-scale perturbation data [16] with STATE makes it uniquely suited to host this project. With the STATE team, we learn how a research group at this scale designs tools, with access to unpublished negative results and design choices that shaped the current architecture.

Timeline & Expected Results

- **Months 1–4:** Train continuous encoder; validate additive composition; build multi-objective optimizer (Aim #1).
- **Months 5–8:** Release Aim #1 code + preprint; curate combination datasets; train interaction encoder (Aim #2).
- **Months 9–12:** Benchmark single-cell non-additive vs. additive prediction; release Aim #2 code + preprint.

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